

HW4

- **Submission date: 03/07/2025 (11:59 PM)**
 - Read your class notes *thoroughly* before attempting the assignment.
 - You can submit **well commented** code in either `.py` or `.ipynb` formats.
 - Submitting a PDF explaining your code is *highly* encouraged.
1. **Scikit-learn tutorial.** The TA has prepared a detailed tutorial explaining the basics of Data Analysis with Scikit-learn using Linear Regression as an example. The tutorial also discusses cross validation. Go through it and try to *reproduce the results on your own*. This exercise will be very helpful for the next few homeworks.
 - For the following questions, you may use calculators, but not programming (use NumPy/Python only for verification. Do not submit that code).
 - **Textbook for this homework:** Lipschutz, S. and Lipson, M., 2001. *Schaum's outline of theory and problems of Probability*. An online copy is available on BU's library website. This book has been reserved for this course. If you cannot access it, please email Tracy Stone (tstone@bu.edu), or Sarah Goodwin (circulation@bu.libanswers.com) for help.
 2. **Conditional Probability.** Suppose a couple has two children. The sample space is $S = \{bb, bg, gb, gg\}$ where we assume an equiprobable space, that is, we assume probability $1/4$ for each point. Find the probability p that both children are boys if it is known that: (a) At least one of the children is a boy. (b) The older child is a boy.
 3. **Independent events.** The probability that A hits a target is $\frac{1}{4}$ and the probability that B hits the target is $\frac{2}{5}$. Both shoot at the target. Find the probability that at least one of them hits the target; that is, find the probability that A or B (or both) hit the target.
 4. **Pairwise Independence vs Independence.** A fair coin is tossed twice. Assume the two tosses are independent. The sample space for this random experiment is $S = \{HH, HT, TH, TT\}$. Define three events
 - $A = \text{"the first toss shows Head"}$
 - $B = \text{"the second toss shows Head"}$
 - $C = \text{"both tosses show the same outcome"}$
 - a. Calculate $\mathbb{P}[A \cap B]$, $\mathbb{P}[A \cap C]$, and $\mathbb{P}[B \cap C]$. What can you say about the pairs of events (A, B) , (A, C) , and (B, C) ?
 - b. Now calculate $\mathbb{P}[A \cap B \cap C]$ and compare with $\mathbb{P}[A] \times \mathbb{P}[B] \times \mathbb{P}[C]$. What does this comparison tell you?
 5. **Random Variables.** Solve problems 5.86, 5.88, and 5.89 in the textbook. Full points will be given only if you show all intermediate steps clearly (this includes problems on plotting CDFs).
 6. Let S be the sample space when a pair of fair dice is tossed. Let X and Y be random variables on S , where $X(a, b) = \max(a, b)$, and Y denotes the sum of the numbers, $Y(a, b) = a + b$. (a) Find the PMF $f_X()$ of X . (b) Find the PMF $f_Y()$ of Y . Also, find the mean (expected) values $\mathbb{E}X$ and $\mathbb{E}Y$.
 7. **Bernoulli and Binomial Distributions.** The probability that Ann hits a target at any time is $p = \frac{1}{3}$. Suppose she fires at the target 7 times. This is a Binomial experiment with $p = \frac{1}{3}, n = 7$. Find the probability that she hits the target: (a) exactly 3 times, and (b) at least 1 time.
 8. Solve problems 6.51, 6.52, 6.61 (a), and 6.63 (expected number E only) in the textbook.

9. **The Gaussian distribution.** Read Sec. 6.4 and Example 6.5 in the textbook before answering this question. Answer questions 6.67 and 6.68 in the textbook.
10. **SciPy stats module.** The `stats` module in SciPy allows users to sample from many continuous distributions. [A full list can be found here](#). We've already seen the normal (Gaussian) distribution in class. How will you sample a set of $N = 10,000$ random numbers in SciPy from (i) the normal distribution, (ii) the exponential distribution, and (iii) the Poisson distribution?
- a) Give the expression for the probability density/mass functions for each of these three distributions.
 - b) Using the N samples, plot the histograms of each of the three distributions.

Hint: Chapter 15 in “The Scientific Python Lectures,” available under “Syllabus and General Information” on the course website explains how to use the `stats` module.